

Lab 5 Preprocessing Heatmaps and Legends

2026-01-16

Lab 5 Heatmaps Preprocessing

Goal

Create a calendar-style heatmap by aggregating total distance by Year and Week.

Importing

```
library(dplyr)
library(tidyr) #recommended for this week

file_id <- "1ymbNqfv9s6YGZzN93HFKAhjg0Z5xZXV1"
url <- paste0("https://drive.google.com/uc?id=", file_id)

df <- read.csv(url)
```

1. Extract required fields

Our objective is to create a checkerboard-style heatmap.
A heatmap typically requires three variables:

- x-axis: Week number
- y-axis: Year
- z-axis (colour): Distance

The first step is to collate the required information into a DataFrame containing:

- Year
- Week
- Distance

Notes:

%G = ISO week-based year

%V = ISO week number (01–53)

```
##   Year Week Distance_km
## 1 2025  30      8.7020
## 2 2025  29      8.3593
## 3 2025  29      8.3725
## 4 2025  29      6.8200
## 5 2025  28      2.7581
## 6 2025  28      5.5963
```

2. Summarise by year and day.

Get the total distance for each week of each year. At this step it is recommended to fill missing values with zero, this will help with general tidyness of your heatmap. A great library for this is `tidyr` imported above. A great function to use is `complete()`.

<https://tidyr.tidyverse.org/reference/complete.html>

3. Save